



Disaggregating confusion? The EU Taxonomy and its relation to ESG rating[☆]

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ABSTRACT

This paper discusses the EU Taxonomy in the context of Environmental, Social, and Governance (ESG) ratings. ESG firm-level ratings tend to differ across ESG data providers, affecting investment decisions due to uncertainty about a firm's sustainability performance. We argue that the EU Taxonomy can support the reduction of this divergence. Using EU Taxonomy-related firm data in tobit regressions, we show that environmental ratings from three out of four ESG data providers are significantly related to the EU Taxonomy. However, our results suggest that the potential for reducing measurement divergence has not yet fully materialised. Our results have implications for investors, ESG data providers, and policy makers.

1. Introduction and literature review

The financial system, as the provider of capital, has been assigned a central role in the sustainable transformation of the global economy (Battiston et al., 2021; Steffen and Schmidt, 2021). Investors increasingly integrate sustainability considerations in decision-making (Shanaev and Ghimire, 2021; Ilhan et al., 2021; Dimson et al., 2020). Thus, information about the sustainability performance of firms is becoming increasingly important. Environmental, social and governance (ESG) ratings are a widely used external source of such information (Dimson et al., 2020; Krueger et al., 2020). However, ESG ratings diverge in the assessment of firms' sustainability performance. In this paper, we analyse how the EU Taxonomy (Taxonomy), a classification system for sustainable economic activities, relates to the environmental part of ESG ratings (E ratings) and discuss how it could help reduce their divergence. Using tobit regressions, we find a significantly positive relation between the Taxonomy and E ratings for three out of four ESG data providers.

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ESG ratings can affect capital allocation and firms' cost of capital, *inter alia*, through changing return expectations (Gibson et al., 2019) and divestment (Krueger et al., 2020). The evolving regulatory landscape could increase the magnitude of those effects. For instance, ESG ratings are likely to become a data source for risk evaluations in the European banking sector (European Banking Authority, 2021; Bundesanstalt für Finanzdienstleistungsaufsicht, 2020) and for determining capital requirements under Basel III / IV (European Banking Authority, 2022; Basel Committee on Banking Supervision, 2021).

However, scholarly work reveals inconsistencies between ESG ratings. Dorfleitner et al. (2015) find an „evident lack of convergence of ESG measures“. Chatterji et al. (2016) show “strong evidence of low commensurability of SRI ratings”. Berg et al. (2019) coin the term *aggregate confusion* based on the weak correlation between ESG ratings. They attribute aggregate confusion to scope divergence (i. e., attributes constituting the definition of sustainability) and measurement divergence (i. e., the indicators utilised to measure a given attribute) (Berg et al., 2019). Those inconsistencies can reinforce effects on capital allocations and firm's cost of capital (Gibson et al., 2019), which may result from uncertainty in a Knightian sense (Knight, 1921).

The European Union (EU) develops the Taxonomy with the objective to harmonise the definition of sustainability and its measurement. The Taxonomy consists of technical screening criteria defining substantial contributions (SC) to the six EU environmental objectives,¹ do no significant harm criteria for the EU environmental objectives, and minimum social safeguards. SC criteria can be understood as the EU's definition of activities supporting the ecological transition of the economy. The criteria should be science-based. So far, the Climate Delegated Act and the Complementary Delegated Act describe technical screening criteria defining SC to climate change mitigation and climate change adaptation, do no significant harm criteria for all six EU environmental objectives and minimum social safeguards (European Commission, 2022, 2021). Definitions of technical screening criteria for SC to the remaining four EU environmental objectives should follow by the end of 2022. The Taxonomy is a key policy initiative within the finance part of the EU Green Deal (European Commission, 2019).

The Taxonomy could help disaggregate the confusion of the environmental part of ESG ratings (E ratings) by harmonising measurement of sustainable economic activities, hence reducing measurement divergence. For this to materialise, E ratings would need to reflect the Taxonomy. Through this transmission channel, the Taxonomy can decrease uncertainty in investment decisions, which ultimately increases market efficiency and reduces firm's average cost of capital. The Taxonomy's central role in the European sustainability ambitions and its informative value should foster this process (Lucarelli et al., 2020). While those effects should mostly unfold in the near future, the E ratings may already reflect the Taxonomy since ESG data providers had access to relevant information during the Taxonomy development process and investors already have the need for Taxonomy-related data in light of the evolving regulatory landscape (Pacces, 2021). Thus, we hypothesise that technical screening criteria for SC to climate change mitigation could have an impact on climate change related aspects in existing E ratings. The Taxonomy could be an explicit part of E rating methodologies or, more likely, implicitly be integrated during the firm's rating process.

In this paper, we demonstrate that a positive relation between the Taxonomy and E ratings already exists for MSCI ESG Research LLC (MSCI), Refinitiv, and V.E, part of Moody's ESG Solutions (V.E). While the relation is not significantly positive for Standard & Poor's (S&P) E ratings. In addition, we find that the relation between Taxonomy and E ratings is stronger pronounced for firms with fully Taxonomy-exposed revenue. To the best of our knowledge, we are the first to use Taxonomy-related firm data in an ESG rating context. Our findings have implications for investors in anticipating financial implications of the Taxonomy, ESG data providers aiming to reduce measurement divergence in E ratings, and policy makers for developing effective sustainable finance regulations. The following section describes the data and methodology. Section 3 discusses the results and Section 4 concludes.

2. Data and methodology

We use a Taxonomy dataset from Institutional Shareholder Services (ISS) ESG that applies the Taxonomy's technical screening criteria to firm revenues. The dataset focuses on SC to climate change mitigation,² i. e., only those revenue-generating activities that substantially contribute to mitigating climate change are further screened for compliance with do no significant harm criteria and minimum social safeguards. The dataset provides information on the firms' share of revenue complying with technical screening criteria for SC to climate change mitigation (SC Alignment) and the firms' share of revenue complying with technical screening criteria for SC, do no significant harm, and minimum social safeguards, see Fig. 1. The dataset is among the first of its kind and we are the first to use it in an ESG rating context.

We analyse the relation between the SC Alignment and E ratings from MSCI, Refinitiv, S&P, and V.E. These E ratings measure a firm's environmental sustainability, i. e., they primarily follow an inside-out perspective. This is in line with the impact perspective of

¹ The six EU environmental objectives are climate change mitigation, climate change adaptation, sustainable use and protection of water and marine resources, transition to a circular economy, pollution prevention and control, and protection and restoration of biodiversity and ecosystems.

² To identify the degree of alignment with defined taxonomy activities, ISS ESG uses approximately 850 Taxonomy-specific screening factors in the following five-step screening process: (1) activity identification, (2) revenue calculation, (3) assessing alignment with screening criteria for substantial contribution, (4) assessing alignment with do no significant harm criteria, and (5) assessing alignment with minimum social safeguard (ISS ESG 2021). The dataset does not reflect the criteria set forth in the Complementary Delegated Act.

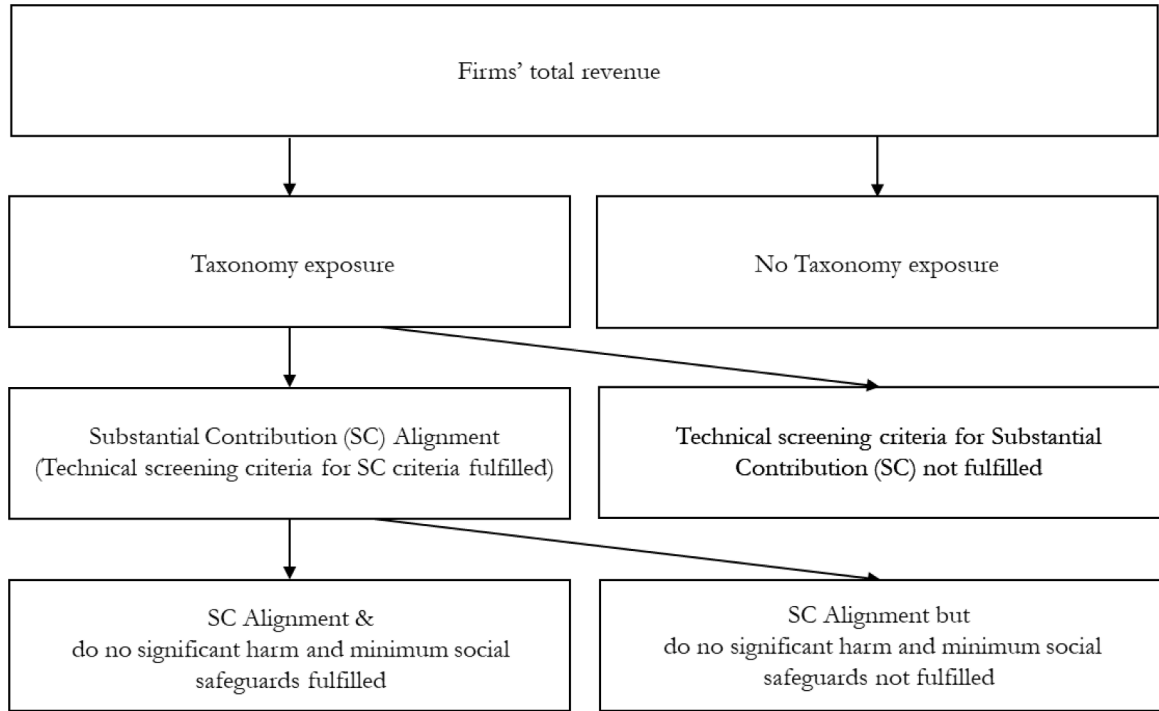


Fig. 1. Firm revenue split by level of alignment with the Taxonomy.

Table 1
Summary statistics.

	N	Mean	SD	Median	0.25	0.75	Min	Max
MSCI E Rating	1566	52.08	18.47	51	39	64	0	100
S&P E Rating	1722	40.66	28.29	34	17	62	0	100
Refinitiv E Rating	1721	50.30	27.28	53.94	28.70	72.48	0	99.05
V. E Rating	1126	34.82	16.93	34	22	46	0	84
Taxonomy Exposure	1813	54.15	39.41	55.65	12	99.98	0	100
Relative SC Alignment	1813	13.80	26.63	0	0	20	0	100
ln(Market Capitalisation)	1809	9.46	2.72	9.03	7.50	11.07	2.54	19.04

Note: This table provides the summary statistics for the used variables. The following variables are presented: total number of observations (N), the mean, the standard deviation (SD), the median, the 25 percent quartile (0.25), the 75 percent quartile (0.75), the minimum (Min) and the maximum (Max).

the Taxonomy, i.e. measuring SC to an environmental objective.³ E ratings substantially reflect climate change related aspects according to their methodology papers (Refinitiv, 2021; SandP 2021; MSCI 2020; V.E 2020). To ensure comparability, we standardise the ratings between 0 and 100. E ratings provided by Refinitiv and V.E do not fully utilise the range in our dataset, see Table 1.

In our analysis, we consider firms with Taxonomy-exposed revenue, yielding a dataset of 1813 firms. E ratings and firm level data are retrieved from Bloomberg, MSCI, Refinitiv, and V.E databases for the years 2019 and 2020⁴ and are matched to the Taxonomy dataset. Rating providers included in our analysis do not cover the full range of all the firms in the ISS ESG Taxonomy dataset. Table 1 provides summary statistics of the dataset.

³ Following this approach, we exclude E ratings that measure different environmental aspects such as the degree of disclosure by a firm (e.g., Bloomberg Environmental Disclosure Score) and the firm's exposure to environmental risks, i.e. the outside-in perspective (e.g., Sustainalytics ESG Risk Rating). Those E ratings provide information on distinctively different questions and do not follow the impact perspective of the Taxonomy, leading to insufficient comparability from a methodological standpoint (Bloomberg, 2022; Sustainalytics, 2021).

⁴ The 2020 data represents the primary data for the analysis. We supplemented 2020 data with 2019 data in cases where 2020 data is missing. The correlation is very high between the yearly datasets ($r > 0.9$) indicating that this approach is appropriate. We acknowledge the effect that the Covid-19 pandemic had on ESG markets (Broadstock et al., 2021; Díaz et al., 2021; Ferriani and Natoli, 2021). Our dataset contains E ratings, for which the data were collected pre-pandemic. Thus, effects of the pandemic should be negligible in this study.

The firm's share of Taxonomy-exposed revenue determines the maximum SC Alignment. This might distort the models since a firm that has maximised its SC Alignment within a low share of Taxonomy-exposed revenue is treated the same as a firm with the same SC Alignment within a higher share of Taxonomy-exposed revenue. We control for this effect by applying the Relative SC Alignment, which is calculated by dividing the firms SC Alignment by its Taxonomy-exposed revenue, see Equation I. Multiplying the result by 100 yields a Relative SC Alignment ranging between 0 and 100.

$$\text{Relative SC Alignment}_i = \frac{\text{SC Alignment}_i}{\text{Firm's Taxonomy - exposed revenue}} \times 100 \quad (1)$$

We follow three steps to analyse how the Taxonomy, as the central and regulatory framework for measuring and assessing SC to climate change mitigation, relates to E ratings in order to determine its potential for reducing measurement divergence. Firstly, we generate the correlation matrix to gain a first indication of the relation between E ratings and Relative SC Alignment. Secondly, since the standardised E ratings range from 0 to 100, we run a tobit regression model on the cross-section of E ratings. We control for the firm characteristics size, domicile, and industry to prevent biased estimates. Size bias should persist for Taxonomy assessment by third party providers as it represents another form of ESG assessment (Drempetic et al., 2020). The Taxonomy cross-references EU legislation, which should improve the Taxonomy's ease of applicability in the EU compared to the global average. In addition, firms domiciled in countries with more ambitious climate change policies should have higher Relative SC Alignments (Gyöngyörová et al., 2021). The level of difficulty in fulfilling the Taxonomy's technical screening criteria differs between industries leading to the need to control for it.

We apply the following tobit regression model (Model I):

$$E \text{ rating}_i = \alpha_1 + \beta_1 \text{Relative SC Alignment}_i + \gamma' \text{Controls}_i + \varepsilon_i \quad (2)$$

where $E \text{ rating}_i$ is the environmental rating of firm i , $\text{Relative SC Alignment}_i$ is firm i 's SC Alignment relative to its Taxonomy-exposed revenue (see Equation I), and Controls_i capture the control variables, i.e., firm size, domicile, and industry classification (NACE).

Thirdly, we expand Model I by further including a dummy variable (Full Taxonomy Exposure) indicating full Taxonomy-exposed revenue and an interaction term between Relative SC Alignment and Full Taxonomy Exposure (Model II). 440 firms have revenue that can be completely mapped to the Taxonomy, representing 24.27% of the full sample. The introduction of the dummy variable *Full Taxonomy Exposure* should correct for the fact that ESG data providers assess the entire firm while we analyse the Relative SC Alignment. Assuming the E ratings and the Taxonomy are related, we would expect that the relation between Relative SC Alignment with the E ratings to be higher on average for firms with full Taxonomy exposure.⁵

3. Findings and discussion

Our analysis suggests that aggregate confusion persists. The correlations between the E ratings by MSCI, S&P, Refinitiv, and V.E range from 0.29 to 0.65 (see Table 2). In line with findings by, inter alia, Chatterji et al. (2016) and Dorfleitner et al. (2015), correlations are significant but do not uniformly show the same level of environmental performance. The size bias in the E ratings highlighted by Drempetic et al. (2020) is significant as well.

The E ratings from MSCI, Refinitiv, S&P and V.E significantly positively correlate with the Relative SC Alignment, indicating a reflection of SC criteria in the E ratings. The magnitude of the correlations differs between ESG data providers. E ratings from S&P and Refinitiv are significantly negatively correlated with the firm's share of Taxonomy-exposed revenue. This finding is in line with expectations since the current coverage of the Taxonomy focuses on emission-intensive industries, meaning that a higher share of Taxonomy-exposed revenue indicates higher revenue generation in these industries. Firm size is negatively correlated to a firm's share of revenue exposed to the Taxonomy. Thus, larger firms seem to be less exposed to the Taxonomy.

Table 2
Correlation matrix.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) MSCI E Rating	1.00						
(2) S&P E Rating	0.29***	1.00					
(3) Refinitiv E Rating	0.29***	0.65***	1.00				
(4) V. E Rating	0.39***	0.60***	0.59***	1.00			
(5) Taxonomy Exposure	0.03	-0.14***	-0.11***	0.04	1.00		
(6) Relative SC Alignment	0.15***	0.07***	0.18***	0.19***	-0.05**	1.00	
(7) ln(Market Capitalisation)	0.07***	0.34***	0.35***	0.05*	-0.21***	0.05**	1.00

Note: This table presents the Bravais-Pearson pairwise correlation coefficients between the E ratings of the four data providers, Taxonomy Exposure, relative SC Alignment and logarithmised market capitalisation. ***, **, and * denote statistical significance at the 1, 5, and 10% levels, respectively.

⁵ For a detailed description of all variables, see Table A.1 in the appendix.

Table 3
Regression results of model I.

	(1) MSCI E Rating	(2) S&P E Rating	(3) Refinitiv E Rating	(4) V. E Rating
Relative SC Alignment	0.105*** (0.017)	0.031 (0.027)	0.125*** (0.022)	0.066*** (0.016)
ln(Market Capitalisation)	0.989*** (0.212)	3.979*** (0.317)	4.116*** (0.295)	1.000*** (0.197)
Domicile Dummies				
EU27	9.230*** (1.319)	18.730*** (1.987)	20.112*** (1.742)	16.265*** (1.451)
North America	-0.033 (1.160)	-2.873* (1.702)	-1.786 (1.629)	-2.977** (1.185)
Rest of the World	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
Constant	24.631*** (3.464)	-8.596** (3.672)	3.626 (4.361)	13.461*** (2.524)
Observations	1563	1719	1718	1124

Note: This table shows the results of the tobit regression model. The dependant variables are the E ratings from the four ESG data providers (columns 1 - 4). The independent variable is the Relative SC Alignment, which measures the SC Alignment in relation to the Taxonomy-exposed revenue of a firm, see Equation I. We control for firm characteristics that have been shown to affect E Ratings, namely logarithmised market capitalisation, domicile dummies (EU27; North America and Rest of the World), as well as industry classification dummies (NACE). For reasons of readability, we do not display the coefficients of all NACE industry classification dummies. Robust standard errors as per [Huber \(1967\)](#) and [White \(1980\)](#) are presented in parentheses. *, **, *** denote statistical significance at 10%, 5% and 1%, respectively.

Table 4
Regression results of model II.

	(1) MSCI E Rating	(2) S&P E Rating	(3) Refinitiv E Rating	(4) V.E E Rating
Relative SC Alignment	0.080*** (0.017)	0.021 (0.030)	0.113*** (0.024)	0.052*** (0.016)
Full Taxonomy Exposure	-2.181 (1.370)	-7.051*** (2.141)	-6.849*** (2.223)	-3.272 (2.032)
Full Taxonomy Exposure x Relative SC Alignment	0.164*** (0.046)	0.090 (0.066)	0.110* (0.057)	0.138** (0.063)
ln(Market Capitalisation)	0.981*** (0.213)	3.879*** (0.320)	4.036*** (0.296)	0.995*** (0.198)
Domicile Dummies				
EU27	9.161*** (1.314)	18.603*** (1.982)	19.996*** (1.741)	16.109*** (1.444)
North America	-0.111 (1.152)	-2.893* (1.703)	-1.757 (1.633)	-3.062** (1.187)
Rest of the World	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
Constant	25.123*** (3.534)	-6.688* (3.706)	5.259 (4.506)	13.509*** (2.533)
Observations	1563	1719	1718	1124

Note: This table presents the results of the tobit regression model. The dependant variables are the E ratings from the four ESG data providers (columns 1 - 4). The independent variable is the Relative SC Alignment, which measures the SC Alignment in relation to the Taxonomy-exposed revenue share of a firm, see Equation 1. In addition to Model I, we consider Full Taxonomy Exposure and the interaction between Full Taxonomy Exposure and Relative SC Alignment. Full Taxonomy Exposure takes a value of 1 if a firm's share of revenue is fully Taxonomy-exposed and 0 otherwise. Including this interaction term corrects for the fact that ESG data providers assess the entire firm while we analyse Relative SC Alignment. As in Model 1, we control for firm characteristics that have been shown to affect E Ratings, namely logarithmised market capitalisation, domicile dummies (EU27; North America and Rest of the World), as well as industry classification dummies (NACE). For reasons of readability, we do not display the coefficients of all NACE industry classification dummies. Robust standard errors as per [Huber \(1967\)](#) and [White \(1980\)](#) are presented in parentheses. *, **, *** denote statistical significance at 10%, 5% and 1%, respectively.

The results from Model I confirm the findings from the correlation analysis, see [Table 3](#). Our results suggest a significantly positive relation of Relative SC Alignment and E ratings from MSCI, Refinitiv, and V.E. Contrary to the correlation matrix, we cannot confirm a significant relation of Relative SC Alignment and E ratings from S&P. We ascribe this effect to the inclusion of control variables, which correct for unobserved bias in the correlation coefficients. Other factors such as domicile, industry, or firm size have significant explanatory value for E ratings as well.

Model II confirms the hypothesis of a positive relation between the level of Taxonomy-exposed revenue and the effect of the Taxonomy on E ratings. The Taxonomy has a stronger explanatory value for E ratings of fully Taxonomy-exposed firms. This finding could indicate that the Taxonomy has a higher potential for reduction of measurement divergence for firms with highly Taxonomy-exposed revenue. We observe this effect for E ratings from MSCI, Refinitiv, and S&P, i.e., the respective interaction terms show a significantly positive relationship, see [Table 4](#). In addition, we find that the Taxonomy retains a significantly positive relation with E ratings from those ESG data providers for the remaining sample. Interestingly, the Taxonomy does not show a significantly positive relation with E ratings from S&P even for fully Taxonomy-exposed firms.

The Taxonomy was not in place when ESG data providers developed the E ratings utilised in our analysis. Nevertheless, we find a positive relation between E ratings and Relative SC Alignment and stronger effects for firms with fully Taxonomy-exposed revenue. Our findings suggest that a potential of the Taxonomy to reduce measurement divergence of E ratings exists, which could reduce greenwashing risks in the financial system. This potential could also exist for social aspects of ESG ratings as well, which could further reduce measurement divergence of ESG ratings. The Platform on Sustainable Finance has already published recommendations on the structure of an EU Social Taxonomy, which the European Commission still needs to translate into legislation ([Platform on Sustainable Finance, 2022](#)). The non-significant relation between the Taxonomy and S&P E ratings suggests that the potential for the reduction in measurement divergence has not fully materialised yet. Finally, the stronger relation for firms with highly Taxonomy-exposed revenue might indicate that the Taxonomy could have the strongest effect on reducing measurement divergence amongst E ratings of pure players and less diversified companies.

Our findings are subject to limitations. Firstly, a time series is necessary to assess a potential reduction in divergence of E ratings once firms have reported Taxonomy data for several years. Secondly, three out of four E ratings show a significant relation to the Taxonomy, leaving some uncertainty as to whether and how the entire ESG rating market might be utilising the Taxonomy. Thirdly, the Taxonomy data used in this paper was generated by a third party, which might not fully align with the Taxonomy data that firms will ultimately report. Fourthly, the current development stage of the Taxonomy includes primarily climate change aspects. Future policy developments will allow advancing this paper's focus as soon as the technical screening criteria for all six EU environmental objectives are available. At this point, the Taxonomy's technical screening criteria should closely resemble indicators currently used in E ratings.

Future research might help to overcome these limitations by analysing a time series once it becomes available (potentially via the European Single Access Point), testing the Taxonomy's effect on other parts of the ESG rating market (e.g., on risk-related ratings), and by running a similar study to ours once the Taxonomy is fully developed. In addition, further research could incorporate components of financial performance to these findings, including the Taxonomy's direct effects on capital allocation and cost of capital. For instance, financial effects of first public disclosure by firms on the Taxonomy may be of interest to the accounting literature. Other research in the field of ESG ratings and sustainability measurement might utilise our perspective on the Taxonomy as a measure to disaggregate confusion of ESG ratings for their purposes.

4. Conclusion

This paper shows the significantly positive relation between E ratings and firm level Taxonomy performance. Our findings are relevant for practitioners, and policy makers: Investors might anticipate potential implications of the Taxonomy on their investments such as changes in cost of capital or capital reallocations by other investors. ESG data providers can use our findings in the process of reducing measurement divergence in E ratings to strengthen reliability and credibility as well as to pre-empt regulation. Policy makers might build on our findings to develop effective sustainable finance regulation that improves the quality of ESG information in the market.

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Appendix

Table A1.

Table A1

Overview of variables.

Variable	Description	Source
<i>Dependent variables</i>		
MSCI E Rating	MSCI E rating measures a firm's environmental performance. Scores range from 0 to 10. We standardized the ratings between 0 and 100 for the purpose of our analysis.	MSCI
S&P E Rating	S&P E rating measures a firm's environmental performance. Scores range from 0 to 100.	Bloomberg
Refinitiv E Rating	Refinitiv E rating measures a firm's environmental performance. Scores range from 0 to 100.	Refinitiv
V. E Rating	V. E rating measures a firm's environmental performance. Scores range from 0 to 100.	Vigeo Eiris
<i>Independent variables</i>		
Taxonomy Exposure	Taxonomy Exposure measures a firm's share of revenue exposed to the EU Taxonomy.	ISS ESG
SC Alignment	SC Alignment measures a firm's share of revenue complying with technical screening criteria for SC to climate change mitigation.	ISS ESG
Relative SC Alignment	Relative SC Alignment is defined as a firm's SC Alignment divided by its share of revenue exposed to the EU Taxonomy multiplied by 100. (+)	ISS ESG
Full Taxonomy Exposure	Full Taxonomy Exposure is a dummy variable based on firms' Taxonomy Exposure. It takes a value of 1 if a firm's revenue exposed to the EU Taxonomy equals 100%, and 0 otherwise. (-)	ISS ESG
<i>Control variables</i>		
Domicile	We introduce three dummy variables, namely EU27, North America and Rest of the World, indicating the firms' domicile.	Refinitiv
Industry classification	We introduce a dummy variable for each industry sector of the NACE classification system.	Refinitiv
ln(Market Capitalisation)	The variable captures a firm's logarithmised market capitalization.	Refinitiv

Note: This table provides a description and the source of the variables used. The expected directions of the regression coefficients are indicated in parentheses.

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